

Schedule Problem

The Passive Warning System (Alerting the Student)

Travel Time Indicator: Display the precise calculated walking time (e.g., "12 min") right next to the passing period time (e.g., "10 min") when a scheduling conflict exists..

Color-Coded Conflict: Use red text or a hazard icon next to the course if the required travel time exceeds the official passing period.

Proximity Buffer Setting: Allow the student to set a personal buffer (e.g., "I need at least 5 minutes extra") that the system factors into its warning calculations.

Building-to-Building Distance Data: Show a simple text summary: "Course A is 1,200 meters from Course B."

The Visual Map Interface

Interactive Route Display: Plot a visual, optimized walking path between the two classroom locations on a campus map pop-up.

Zoomable Floor Plans: For multi-story buildings, allow the student to zoom in to see the floor of the classroom, accounting for time spent using stairs or elevators.

Landmark Reference: Display key campus landmarks (e.g., "Passes the Library," "Near the Student Union") along the calculated route for easy identification.

The Optimal Schedule Recommender

Proximity Scoring: Assign an internal "Proximity Score" (e.g., A+ to F) to each potential schedule, based on the total daily travel time required.

Section Filtering: Add a filter to the course catalog that allows the student to "Filter by Closest Section" to their previous class selection.

Conflict Resolver Button: A one-click button that automatically swaps a conflicting course section for the next nearest available section that doesn't violate the travel time rule.

Block Scheduling Suggestions: Suggest grouping all classes on one side of campus on certain days to minimize daily travel (e.g., "Monday/Wednesday is North Campus Day").

The Post-Registration Planner

Personalized Itinerary: Generate a daily schedule that lists the time a student must depart the first class to arrive on time for the second.

Route Sharing: Allow students to share their specific transition route and travel time data with peers taking the same back-to-back classes.

Real-Time Bus/Shuttle Integration: If applicable, incorporate real-time GPS data for campus shuttles, calculating if using transit is faster than walking.

Quickest Path Finder: A simple mobile tool where a student can quickly enter two buildings and receive the optimal route and travel time.